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(54) **PROJECTION OBJECTIVE**

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(57) **ABSTRACT**

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Related U.S. Application Data

(63) Continuation-in-part of application No. PCT/EP01/14846, filed on Dec. 15, 2001.

Projection exposure device and also projection objective with a lens arrangement which has at least one lens group of negative refractive power, this lens group comprising at least four lenses of negative refractive power, and a lens of positive refractive power being arranged after the third lens of negative refractive power in this lens group.

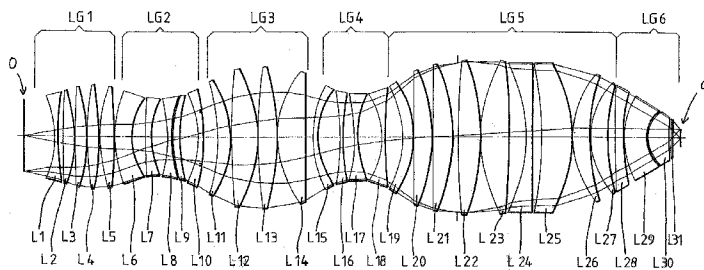
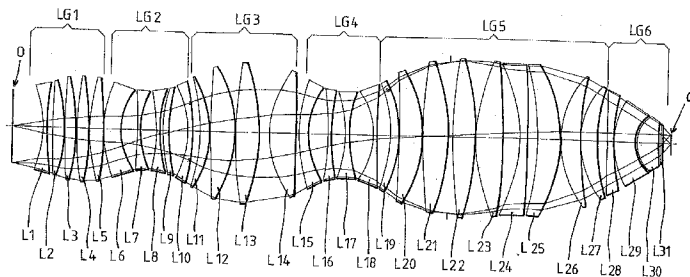


FIG. 1

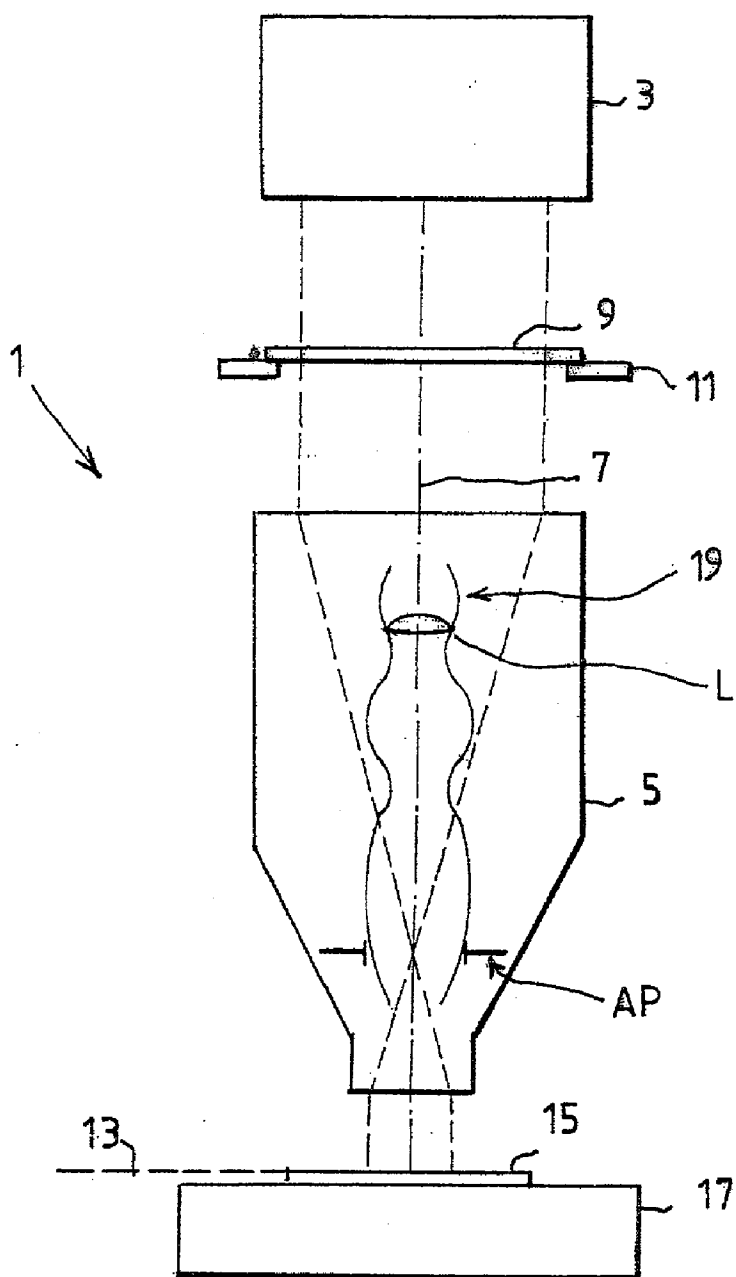


FIG. 2

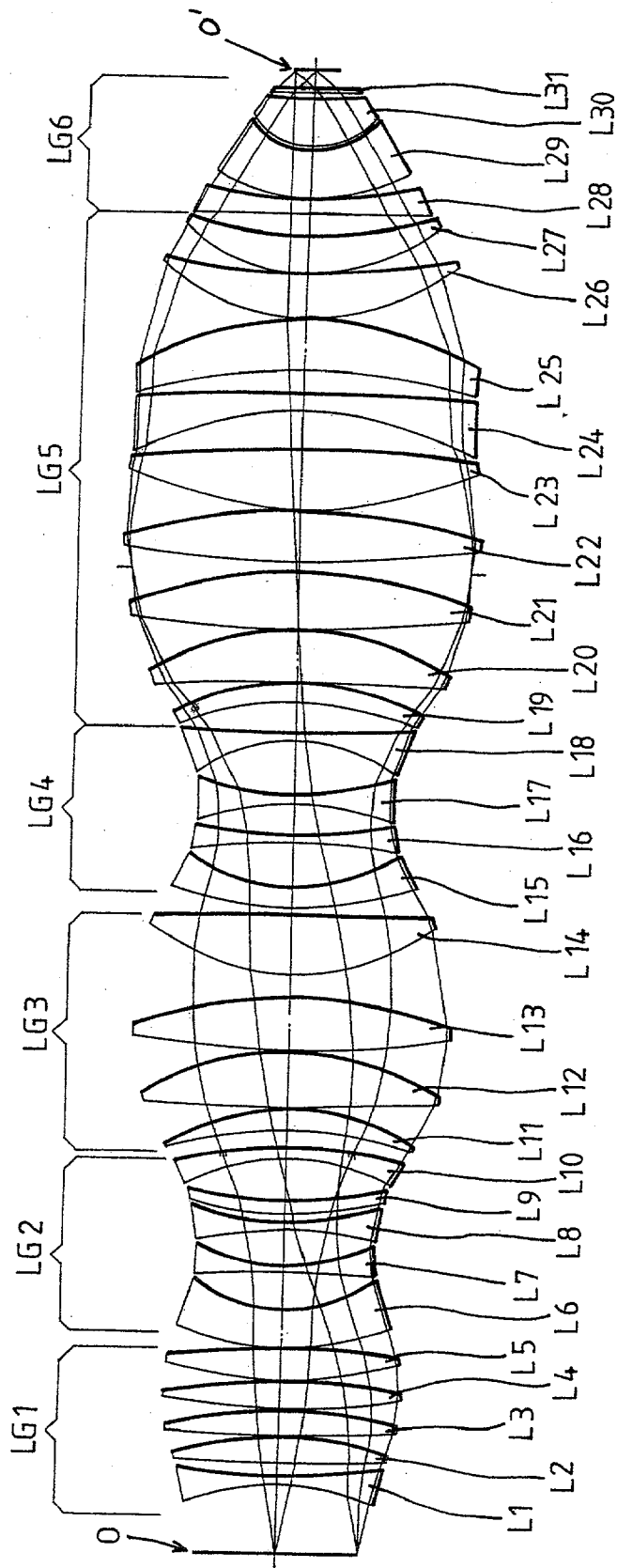


FIG. 3

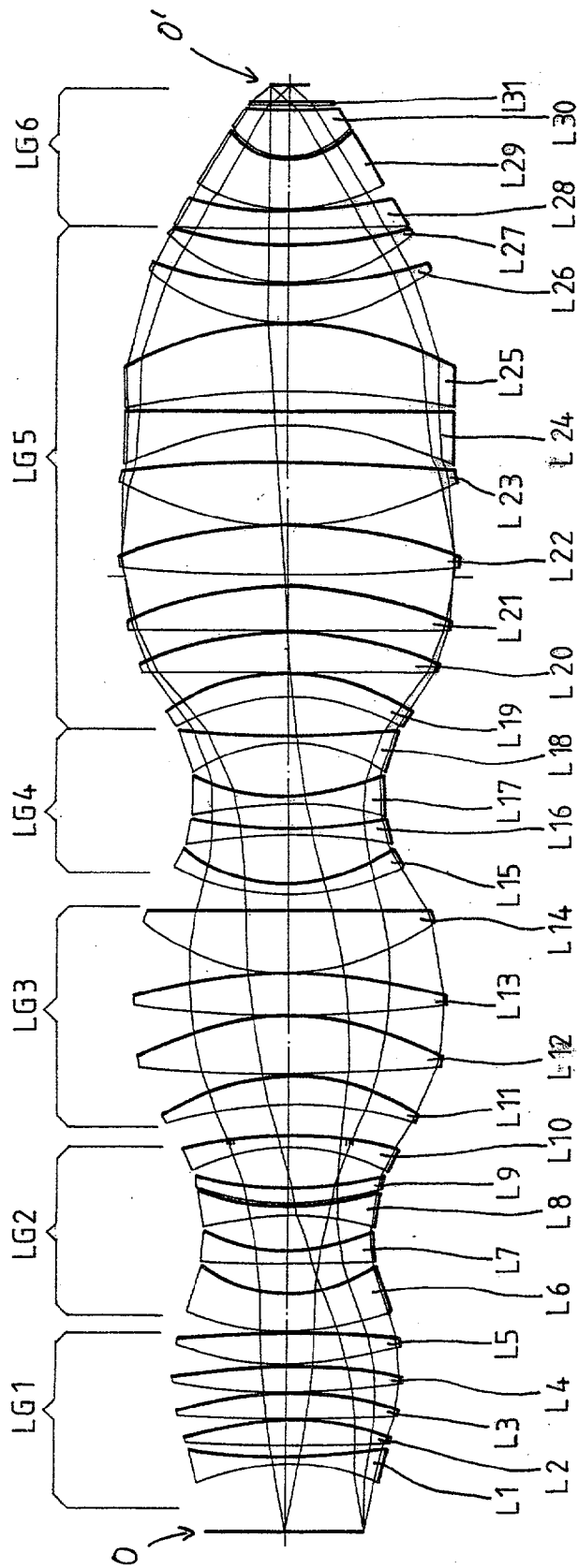


FIG. 4

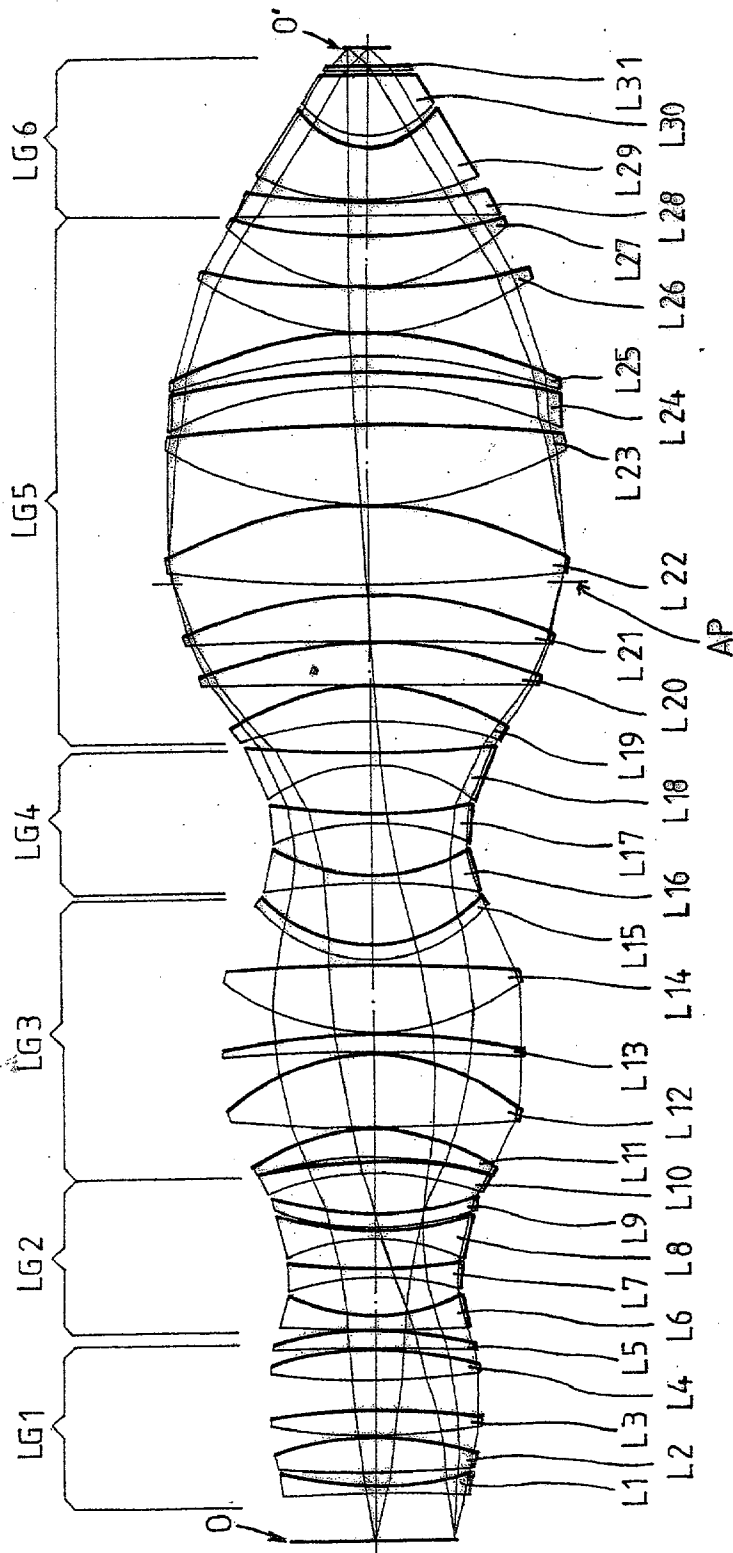
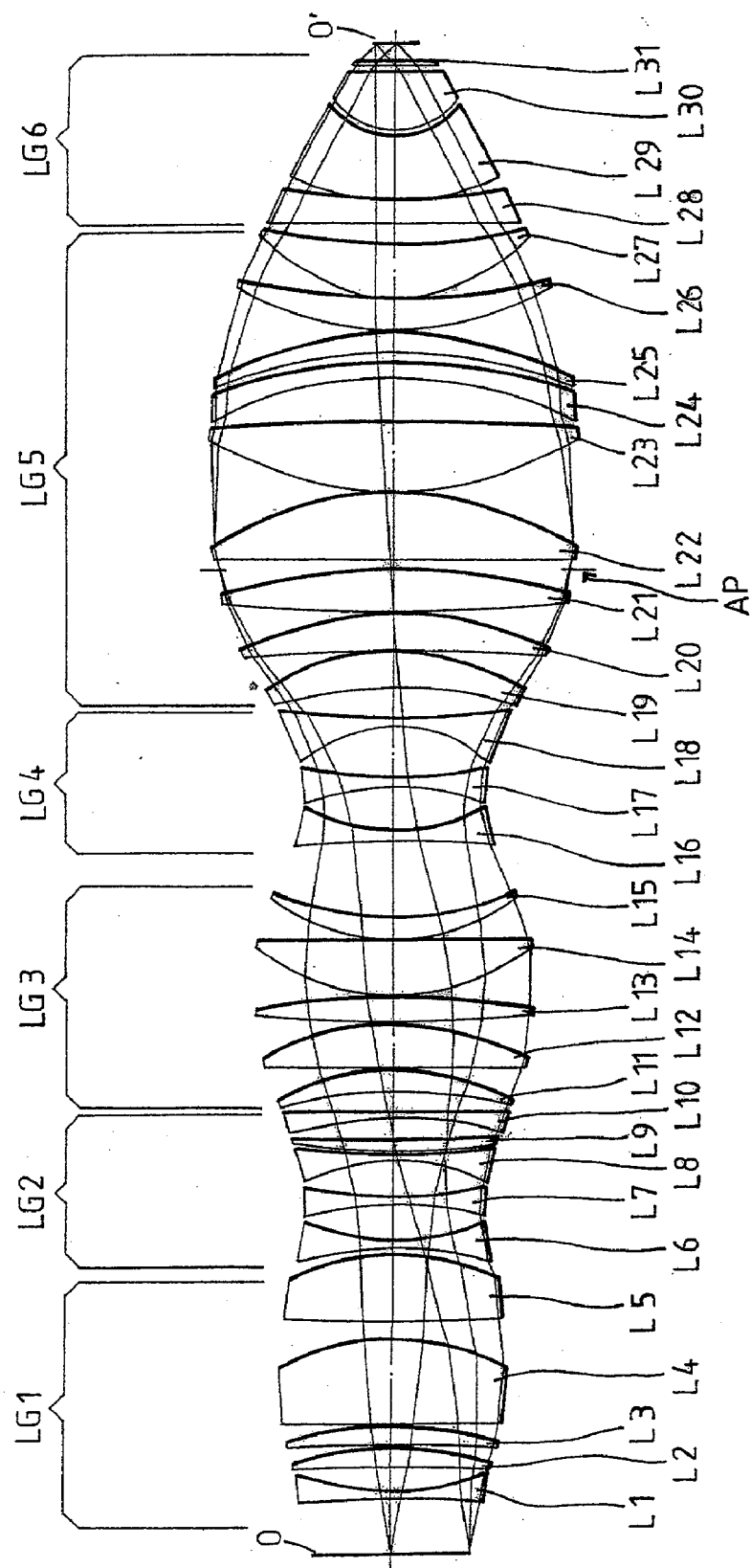


FIG. 5



PROJECTION OBJECTIVE

CROSS-REFERENCES TO RELATED APPLICATIONS

[0001] This application is a continuation application of PCT/EP01/14846, which is pending.

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

[0002] Not applicable.

BACKGROUND OF THE INVENTION

[0003] The invention relates to a projection objective having at least one lens group of negative refractive power, this lens group including at least four lenses of negative refractive power.

TECHNICAL FIELD

[0004] From German Patent Document DE 199 42 281.8, FIGS. 8-10, a projection objective is known whose first lens group with negative refractive power consists of four negative lenses. Projection objectives are likewise known from European Patent Document EP 712 019 A2, U.S. Pat. No. 5,969,803, U.S. Pat. No. 5,986,824 and DE 198 18 444 A1 with a first lens group of negative refractive power, consisting of at least four negative lenses.

[0005] A projection objective is known from U.S. Pat. No. 5,990,926 and has a first lens group of negative refracting power, consisting of three negative lenses. Projection objectives are known from the documents U.S. Pat. No. 6,084,723, EP 721 150 A2, U.S. Pat. No. 6,088,171, and DE 198 18 444 A1, and have a lens group of negative refractive power, which forms a first waist and which consists of four negative lenses, a lens of positive refractive power being arranged after the first negative lens.

[0006] From DE 199 42 281, FIGS. 2-4, a lens group of negative refractive power is known which consists of four negative lenses, a positive lens being arranged after the second negative lens. A meniscus lens curved toward the image is provided as the positive lens.

SUMMARY OF THE INVENTION

[0007] The invention has as its object to develop a lens group of negative refractive power which has an advantageous effect on the imaging properties of a projection objective.

[0008] The object of the invention is attained by a lens of positive refractive power arranged in this lens group after a third lens of negative refractive power.

[0009] A further object of the invention is to improve the imaging properties of a projection objective, particularly for an illumination wavelength of 193 nm, with little use of calcium fluoride.

[0010] The imaging properties of the objective could be improved by the measure of constructing a lens group of negative refractive power such that this lens group of negative refractive power consists of four lenses of negative refractive power, a lens of positive refractive power being arranged after the third lens of negative refractive power.

This configuration with the lens of positive refractive power has advantageous effects on astigmatism and coma correction.

[0011] It has also been found to be advantageous to provide a meniscus lens as the positive lens. This additionally makes possible a favorable effect on the sagittal coma.

[0012] It has furthermore been found to be advantageous that the lens of positive refractive power is provided with a convex lens surface on the object side.

[0013] Further advantageous measures are described in the claims.

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] The invention is described in detail using a few embodiment examples.

[0015] FIG. 1 shows a projection exposure device;

[0016] FIG. 2 shows a lithographic objective, particularly for 193 nm;

[0017] FIG. 3 shows a lithographic objective, particularly for the wavelength 193 nm;

[0018] FIG. 4 shows a lithographic objective, particularly for an illumination wavelength of 351 nm;

[0019] FIG. 5 shows a lithographic objective, particularly for an illumination wavelength of 351 nm.

DETAILED DESCRIPTION OF THE INVENTION

[0020] The construction of a projection exposure device is first described in principle using

[0021] FIG. 1. The projection exposure device 1 has an illumination device 3 and a projection objective 5. The projection objective comprises a lens arrangement 19 with an aperture diaphragm AP, an optical axis 7 being defined by the lens arrangement 19. A mask 9 is arranged between the illumination device 3 and the projection objective 5, and is held in the beam path by means of a mask holder 11. Such masks used in microlithography have a micrometer or nanometer structure. This structure of the mask is imaged on an image plane 13, reduced by means of the projection objective 5 by up to a factor of ten, in particular by the factor four. A substrate or a wafer 15, positioned by a substrate holder 17, is held in the image plane 13. The minimum structures which can still be resolved depend on the wavelength λ of the light used for illumination and also on the image-side aperture of the projection objective 5; the maximum attainable resolution of the projection exposure device 1 increases with decreasing wavelength of the illumination device 3 and with increasing aperture of the projection objective 5.

[0022] Different embodiments of lens arrangements 19 are shown in FIGS. 2 and 5.

[0023] FIG. 2 shows a lens arrangement 19 which is designed for an illumination wavelength of 193 nm and an image-side aperture of 0.75. In this lens arrangement, the distance between the projection plane 0 and image plane 0' is 1,000 mm. The projection objective shown comprises 31 lenses L1-L31, which can be divided into six lens groups LG1-LG6.

[0024] A first lens group LG1 has positive refractive power, and consists of the lenses L1-L5. The adjoining lens group LG2 has overall negative refractive power. The first lens L6 of this lens group is a thick meniscus lens, the middle thickness of which in the region of the optical axis is at least 15% of the maximum lens diameter. This lens has a particularly advantageous effect on the flattening of the image surface in the tangential and sagittal directions.

[0025] This lens L6 is followed by two further lenses of negative refractive power. In this embodiment example, two biconcave lenses are provided for these lenses L7 and L8. The adjoining lens L9 has positive refractive power. This lens L9 is a meniscus lens, curved toward the image, with a concave radius of curvature on the image side. An advantageous effect is obtained with this lens with respect to astigmatism, coma, and sagittal coma. The following lens L10 has negative refractive power and is a meniscus lens curved toward the image. This lens L10 is provided with an aspheric surface on the image side. In particular, image errors in the region between the image field zone and the image field margin can be corrected by means of this aspheric. This correction effects an increase of the image quality, particularly in the sagittal direction. A waist is formed by this second lens group LG2 having negative refractive power.

[0026] The adjoining third lens group LG3 has positive refractive power and consists of the lenses L11-L14. A fourth lens group LG4 follows this third lens group LG3 and has negative refractive power, and a second waist is formed by it. This fourth lens group LG4 comprises the lenses

L15-L18, the lens L15 being a meniscus lens curved toward the image, with a concave surface curvature on the image side.

[0027] The fifth lens group LG5 comprises the lenses L19-L27 and has positive refractive power overall. A diaphragm is arranged between the positive lenses L21 and L22. The maximum diameter of this lens group or of the projection objective is about 240 mm. The sixth lens group LG6 likewise has positive refractive power, and comprises the lenses L28-L31, the lens L31 being a plane parallel plate. The lens L30, which is heavily stressed by radiation, consists of calcium fluoride to reduce compaction. Quartz glass is provided as the lens material for the remaining lenses. The use of quartz glass as the lens material has the advantage that this material, in comparison with calcium fluoride, is in the first place available on the market, and also is a less expensive material in comparison with fluoride crystals such as e.g. calcium fluoride and barium fluoride, to name only a few here.

[0028] With an image field of 28.04 mm, the longitudinal color error for the bandwidth of 0.25 pm (thus ± 0.125 pm) is a maximum of 57.5 nm. The transverse color error reaches no greater value than 1.2 nm for $\Delta\lambda \pm 0.125$ pm. The RMS value is an established measure, e.g., established in CODE V, of how strongly the wavefront deviates from the wavefront of an ideal spherical wave. In this embodiment example, the RMS value is smaller than 7.0 mx for all image points. The numerical aperture of this projection objective is 0.75.

[0029] The exact lens data are given in Table 1.

TABLE 1

193.304 nm m1301b				Refractive indices		
Lens Surface	Radii	Thicknesses	Glasses	Hmax	193.304 nm	193.804 nm
1	Infinity	14.6448	L710	62.030	.999982	.999982
2	-166.55059	7.0000	SIO2	62.490	1.560289	1.559505
3	506.05751	7.6109	HE	68.747	.999712	.999712
4	2154.09731	17.8345	SIO2	71.001	1.560289	1.559505
5	-236.44399	.7000	HE	72.626	.999712	.999712
6	1300.07069	17.5997	SIO2	76.997	1.560289	1.559505
7	-300.86909	.7000	HE	77.840	.999712	.999712
8	485.10529	18.6566	SIO2	79.774	1.560289	1.559505
9	-465.03408	.7000	HE	79.824	.999712	.999712
10	269.93230	21.2497	SIO2	78.221	1.560289	1.559505
11	-759.19722	.7000	HE	77.300	.999712	.999712
12	210.59503	26.0777	SIO2	71.714	1.560289	1.559505
13	100.59316	23.6901	HE	60.361	.999712	.999712
14	-1723.36549	6.3000	SIO2	60.021	1.560289	1.559505
15	134.93156	23.1342	HE	58.507	.999712	.999712
16	-253.77701	6.3000	SIO2	58.968	1.560289	1.559505
17	207.65329	3.7881	HE	63.355	.999712	.999712
18	248.43656	10.0000	SIO2	64.814	1.560289	1.559505
19	289.02332	27.2836	HE	66.785	.999712	.999712
20	-149.66622	8.0000	SIO2	68.386	1.560289	1.559505
21	-343.08973A	12.2670	HE	76.791	.999712	.999712
22	-261.71401	14.0893	SIO2	82.489	1.560289	1.559505
23	-158.93251	.7056	HE	84.325	.999712	.999712
24	2089.67024	36.8436	SIO2	99.088	1.560289	1.559505
25	-197.66453	.7000	HE	101.087	.999712	.999712
26	656.07716	36.9267	SIO2	106.566	1.560289	1.559505
27	-310.68503	15.0566	HE	106.873	.999712	.999712
28	163.06222	40.6761	SIO2	96.385	1.560289	1.559505
29	-4111.47456	3.5361	HE	93.567	.999712	.999712
30	250.51145	15.0000	SIO2	82.093	1.560289	1.559505
31	127.80623	29.9608	HE	70.545	.999712	.999712
32	-395.01305	6.3000	SIO2	69.409	1.560289	1.559505

TABLE 1-continued

193.304 nm m1301b					Refractive indices	
Lens Surface	Radii	Thicknesses	Glasses	Hmax	193.304 nm	193.804 nm
33	365.10790	20.3664	HE	65.891	.999712	.999712
34	-193.49584	6.3000	SIO2	65.302	1.560289	1.559505
35	181.51555	36.3017	HE	65.615	.999712	.999712
36	-117.90431	6.3037	SIO2	66.498	1.560289	1.559505
37	1509.29994	19.7322	HE	77.925	.999712	.999712
38	-217.57963	13.5438	SIO2	79.866	1.560289	1.559505
39	-187.32781	.7000	HE	85.009	.999712	.999712
40	-1458.10055	33.4944	SIO2	97.198	1.560289	1.559505
41	-205.38045	.7000	HE	101.387	.999712	.999712
42	975.11627	38.4016	SIO2	113.083	1.560289	1.559505
43	-328.97948	-.1900	HE	114.925	.999712	.999712
44	Infinity	7.0480	HE	116.031	.999712	.999712
	Diaphragm	.0000		116.031		
45	862.15240	33.9785	SIO2	119.182	1.560289	1.559505
46	-429.49378	.7000	HE	119.854	.999712	.999712
47	290.91831	39.9124	SIO2	118.102	1.560289	1.559505
48	-1135.09587	25.2506	HE	116.490	.999712	.999712
49	-250.90293	12.0000	SIO2	115.346	1.560289	1.559505
50	-2698.09888	15.7215	HE	114.878	.999712	.999712
51	-428.51713	35.3166	SIO2	114.769	1.560289	1.559505
52	-229.75028	.7000	HE	116.163	.999712	.999712
53	157.14389	29.4847	SIO2	100.025	1.560289	1.559505
54	407.42757	.7000	HE	97.710	.999712	.999712
55	135.43900	28.6103	SIO2	89.777	1.560289	1.559505
56	312.12527	12.0496	HE	82.745	.999712	.999712
57	1941.44476	12.0000	SIO2	80.391	1.560289	1.559505
58	299.45406	.7000	HE	70.837	.999712	.999712
59	134.94509	33.4425	SIO2	64.789	1.560289	1.559505
60	56.32593	1.3033	HE	42.057	.999712	.999712
61	53.43496	33.3158	CAF2	41.245	1.501436	1.500946
62	353.20173	3.0808	HE	31.464	.999712	.999712
63	Infinity	3.0000	SIO2	29.528	1.560289	1.559505
64	Infinity	12.0000	L710	27.867	.999982	.999982
65	Infinity			14.020*	1.0	1.0

*This height does not have a limiting effect
L710 is air at 950 mbar
Aspheric constants of the surface 21:
K = 0, 10137327 * 10⁺² C₁ = 0, 40139591 * 10⁻⁷ C₂ = 0, 11813452 * 10⁻¹²
C₃ = 0, 45259406 * 10⁻¹⁶ C₄ = -29594696 * 10⁻²⁰ C₅ = 0, 11174027 * 10⁻²³
C₆ = 0, 62923814 * 10⁻²⁸

[0030] A reduction of the transverse color error by about 30% can be attained by the provision of the lens L11 as calcium fluoride and slight modifications of the lenses of the lens arrangement 19. With an image field of 28.04 mm, the

transverse color error is a maximum of ±0.82 nm for λ±0.125 pm in this variant embodiment, and the longitudinal color error is at most ±57.5 nm. The lens data of the modified variant with two calcium fluoride lenses are given in Table 2.

TABLE 2

193.304 nm m1301a					Refractive indices	
Lens Surface	Radii	Thicknesses	Glasses	Hmax	193.304 nm	193.804 nm
1	Infinity	14.5311	L710	62.030	.999982	.999982
2	-168.01710	7.0000	SIO2	62.491	1.560289	1.559505
3	494.67225	7.8655	HE	68.711	.999712	.999712
4	2470.60478	17.5906	SIO2	70.967	1.560289	1.559505
5	-238.16599	.7000	HE	72.602	.999712	.999712
6	1299.55339	17.5821	SIO2	77.012	1.560289	1.559505
7	-301.66419	.7000	HE	77.869	.999712	.999712
8	481.31484	18.6248	SIO2	79.887	1.560289	1.559505
9	-473.57399	.7000	HE	79.949	.999712	.999712
10	266.88043	21.7634	SIO2	78.439	1.560289	1.559505
11	-708.35185	.7000	HE	77.528	.999712	.999712
12	214.11073	26.1927	SIO2	71.854	1.560289	1.559505
13	100.96743	23.5803	HE	60.436	.999712	.999712
14	-1862.02312	6.3000	SIO2	60.094	1.560289	1.559505

TABLE 2-continued

193.304 nm m1301a					Refractive indices	
Lens Surface	Radii	Thicknesses	Glasses	Hmax	193.304 nm	193.804 nm
15	134.04061	23.1274	HE	58.551	.999712	.999712
16	-258.86710	6.3000	SIO2	59.015	1.560289	1.559505
17	209.11797	4.1357	HE	63.332	.999712	.999712
18	260.76414	10.0000	SIO2	64.773	1.560289	1.559505
19	306.31791	26.7167	HE	66.774	.999712	.999712
20	-150.53952	8.0000	SIO2	68.362	1.560289	1.559505
21	-343.08973A	12.2234	HE	76.639	.999712	.999712
22	-255.62602	14.5522	CAF2	82.065	1.501436	1.500946
23	-155.60089	.7135	HE	84.050	.999712	.999712
24	2570.44473	36.4559	SIO2	99.015	1.560289	1.559505
25	-197.34110	.7000	HE	101.069	.999712	.999712
26	658.21211	37.5330	SIO2	106.976	1.560289	1.559505
27	-304.78835	12.5228	HE	107.346	.999712	.999712
28	165.31624	40.9031	SIO2	97.479	1.560289	1.559505
29	-4380.57486	4.7796	HE	94.694	.999712	.999712
30	229.18415	15.0000	SIO2	81.886	1.560289	1.559505
31	127.94022	29.9427	HE	70.764	.999712	.999712
32	-406.54392	6.3000	SIO2	69.593	1.560289	1.559505
33	319.24233	21.1149	HE	65.789	.999712	.999712
34	-194.65176	6.3000	SIO2	65.210	1.560289	1.559505
35	181.88877	36.3561	HE	65.538	.999712	.999712
36	-117.18877	6.3000	SIO2	66.424	1.560289	1.559505
37	1557.90278	19.4997	HE	77.908	.999712	.999712
38	219.93598	13.5189	SIO2	79.860	1.560289	1.559505
39	-188.67644	.7000	HE	84.977	.999712	.999712
40	-1338.43234	33.3397	SIO2	96.879	1.560289	1.559505
41	-203.66807	.7000	HE	101.108	.999712	.999712
42	940.74664	37.9766	SIO2	113.017	1.560289	1.559505
43	-328.34415	.0000	HE	114.751	.999712	.999712
44	Infinity	7.0063	HE	115.860	.999712	.999712
	Diaphragm	.0000		115.860		
45	852.93737	33.9995	SIO2	119.015	1.560289	1.559505
46	-429.06783	.7000	HE	119.681	.999712	.999712
47	293.26753	39.6485	SIO2	117.910	1.560289	1.559505
48	-1122.40839	25.1587	HE	116.310	.999712	.999712
49	-250.45086	12.0000	SIO2	115.174	1.560289	1.559505
50	-2403.16353	15.7564	HE	114.736	.999712	.999712
51	-418.77311	35.6541	SIO2	114.623	1.560289	1.559505
52	-229.51380	.7000	HE	116.090	.999712	.999712
53	157.59158	29.5131	SIO2	100.117	1.560289	1.559505
54	411.55925	.7000	HE	97.819	.999712	.999712
55	135.23398	26.6934	SIO2	86.824	1.560289	1.559505
56	312.28548	12.0796	HE	82.801	.999712	.999712
57	1974.55831	12.0000	SIO2	80.455	1.560289	1.559505
58	304.24095	.7000	HE	70.960	.999712	.999712
59	135.28036	33.4905	SIO2	64.839	1.560289	1.559505
60	56.37313	1.2983	HE	42.062	.999712	.999712
61	53.46309	33.2706	CAF2	41.247	1.501436	1.500946
62	351.07410	3.0893	HE	31.472	.999712	.999712
63	Infinity	3.0000	SIO2	29.536	1.560289	1.559505
64	Infinity	12.0000	L710	27.875	.999982	.999982
65	Infinity			14.020*	1.0	1.0

*This height does not have a limiting effect
L710 is air at 950 mbar
Aspheric constants of the surface 21:
 $K = -0, 10137327 * 10^{+2}$ $C_1 = 0, 40139591 * 10^{-07}$ $C_2 = 0, 11813452 * 10^{-12}$
 $C_3 = 0, 45259406 * 10^{-16}$ $C_4 = -0, 29594696 * 10^{-20}$ $C_5 = 0, 11174027 * 10^{-23}$
 $C_6 = -0, 62923814 * 10^{-28}$

[0031] The lens arrangement 19 shown in FIG. 3 has 31 lenses, L1-L31, which can be divided into six lens groups LG1-LG6. The distance between the object plane 0 and the image plane 0' is 1,000 mm.

[0032] The first lens group has positive refractive power and consists of the lenses L1-L5. The first lens L1 is a biconcave lens and has negative refractive power. The lenses L2-L5 following it are biconvex lenses which have positive refractive power.

[0033] The second lens group LG2 consists of the lenses L6-L10, the lenses L6-L8 having negative refractive power. The lens L9 has positive refractive power. This lens L9 is again a meniscus lens with a concave curved surface on the image side. Lens L10 has negative refractive power and is provided with an aspheric lens surface on the image side. This lens surface in particular permits higher order image errors to be corrected.

[0034] The adjoining lens group LG3 has positive refractive power. A bulge is formed by this lens group with the lenses L11-L14. The lens L14 is provided with a plane surface on the image side. The arrangement of the lens group L3 has the distinctive feature that unusually large air distances are provided on both sides, between lens group LG3 and the lens groups LG2 and LG4 bordering on it. By the special arrangement of the third lens group, the deviation of the wavefront could be reduced over all image heights, in this doubly telecentric objective having a high numerical aperture of 0.75, with little use of aspherics and a constructional length of 00'=1,000 mm. The sum of the two air spaces before and after LG3 is markedly greater than the sum of the glass thicknesses of the following lens group LG4. This has a particularly advantageous effect on the transverse aberrations.

[0035] The fourth lens group, by which a second waist is formed, consists of the lenses L15-L18. The lens L15 is curved toward the object. The lens L19, of the adjoining lens group LG5, has lens surfaces which run nearly parallel and are curved toward the image. The difference of the radii is

smaller than 3% with respect to the smaller radius. In particular, the absolute difference of radii is smaller than 4 mm. The refractive power of this lens L19 is very small, with $f_{19}>4,000$.

[0036] The lens group LG5 includes the further lenses L20-L27, a diaphragm being arranged between the lenses L21 and L22. The last lens group, LG6, is formed by the lenses L28-L31, L31 being a plane parallel plate.

[0037] This lens arrangement 19 shown in FIG. 3 is designed for the wavelength 193 nm. The bandwidth of the light source is 0.25 pm. A field of 10.5×26 mm can be exposed by means of this lens arrangement 19. The numerical aperture of this lens arrangement is 0.75 on the image side. The RMS value, as a deviation from the ideal spherical wave, is monochromatically smaller than 5 mλ with respect to 193 nm. The transverse color error is smaller than ±1.4 nm for Δλ±0.1 25 pm, and the longitudinal color error is smaller than ±58.75 nm in the whole image field.

[0038] The exact lens data are given in Table 3.

TABLE 3

M1650p Lens Surface	Radii	Thicknesses	Glasses	Refractive index at	
				193.304 nm	½ Free Diameter
0	0.000000000	45.969574340	L710	0.99998200	56.080
1	-163.893583594	7.000000000	SIO2	1.56028895	62.298
2	543.276868900	6.914746504	HE	0.99971200	68.527
3	4925.150504439	17.003634307	SIO2	1.56028895	70.200
4	-237.663812578	0.700000000	HE	0.99971200	71.892
5	1399.349047982	18.279497082	SIO2	1.56028895	76.260
6	-282.822296960	0.700000001	HE	0.99971200	77.210
7	485.041831962	19.095677657	SIO2	1.56028895	79.259
8	-448.790019973	0.700000000	HE	0.99971200	79.327
9	261.052873524	21.940801476	SIO2	1.56028895	77.716
10	-776.403478410	0.700000000	HE	0.99971200	76.687
11	210.308913355	25.735549153	SIO2	1.56028895	71.175
12	99.341848189	22.473973608	HE	0.99971200	59.883
13	0.000000000	8.009409065	SIO2	1.56028895	59.565
14	130.189257019	24.765647204	HE	0.99971200	57.743
15	-236.044200760	7.570027862	SIO2	1.56028895	58.334
16	225.745307153	1.721089583	HE	0.99971200	62.796
17	249.257978944	10.000000000	SIO2	1.56028895	63.335
18	290.876194298	26.689502502	HE	0.99971200	65.179
19	-157.850195148	8.000000000	SIO2	1.56028895	67.322
20	-343.089730000A	21.241706809	HE	0.99971200	74.633
21	-300.686931187	19.035466008	SIO2	1.56028895	85.971
22	-166.339801171	1.155814165	HE	0.99971200	88.981
23	1242.960741027	39.709671515	SIO2	1.56028895	103.381
24	-217.561562194	0.700000000	HE	0.99971200	105.365
25	873.041438293	28.702760134	SIO2	1.56028895	108.135
26	-394.309651697	0.700000002	HE	0.99971200	108.176
27	168.490233663	42.038473874	SIO2	1.56028895	100.522
28	0.000000000	13.298461584	HE	0.99971200	97.459
29	198.769354524	7.491654073	SIO2	1.56028895	79.508
30	127.110375082	32.181225955	HE	0.99971200	72.267
31	-412.500785204	6.000000000	SIO2	1.56028895	70.633
32	396.017439811	16.248991558	HE	0.99971200	67.019
33	-267.221433894	6.000000000	SIO2	1.56028895	66.551
34	168.512314408	36.354622161	HE	0.99971200	65.389
35	-122.758597736	6.049883829	SIO2	1.56028895	66.079
36	766.523644100	27.033401060	HE	0.99971200	75.781
37	-159.522000000	16.394944690	SIO2	1.56028895	77.809
38	-155.641000000	0.700000000	HE	0.99971200	84.996
39	0.000000000	27.99555874	SIO2	1.56028895	101.051
40	-266.016736880	0.890470375	HE	0.99971200	103.561
41	0.000000000	30.098513441	SIO2	1.56028895	110.668
42	-290.836485170	7.000000010	HE	0.99971200	112.215

TABLE 3-continued

M1650p Lens Surface	Radii	Thicknesses	Glasses	Refractive index at	
				193.304 nm	½ Free Diameter
43	0.000000000	0.431524267	HE	0.99971200	115.718
44	1323.425726038	33.775984345	SIO2	1.56028895	119.885
45	-354.247417821	0.700000000	HE	0.99971200	120.693
46	253.851238303	43.981502597	SIO2	1.56028895	119.436
47	-1091.287744312	24.113176044	HE	0.99971200	117.882
48	-256.378455601	10.108953957	SIO2	1.56028895	116.975
49	510547.419151742	13.749960972	HE	0.99971200	115.781
50	-549.718956763	46.950737825	SIO2	1.56028895	115.712
51	-237.550046531	0.733009323	HE	0.99971200	117.030
52	152.354106326	27.736146606	SIO2	1.56028895	98.476
53	340.293718782	0.709804301	HE	0.99971200	95.981
54	131.460836161	25.026141744	SIO2	1.56028895	85.578
55	293.107264484	10.404798830	HE	0.99971200	82.205
56	1251.239137872	12.817811407	SIO2	1.56028895	81.001
57	260.342262866	1.378798137	HE	0.99971200	70.436
58	130.810580199	34.683194416	SIO2	1.56028895	64.390
59	55.085948639	0.700000000	HE	0.99971200	41.181
60	52.721068002	33.140446950	CAF2	1.50143563	40.631
61	357.005545933	2.868260812	HE	0.99971200	30.936
62	0.000000000	3.000000000	SIO2	1.56028895	29.212
63	0.000000000	12.000000000	L710	0.99998200	27.572
64	0.000000000	0.000000000		1.00000000	14.020

L710 is air at 950 mbar
Aspheric constants of the surface 20:
K 10.1373
C1 4.01395910e - 008
C2 1.18134520e - 013
C3 4.52594060e - 017
C4 -2.95946970e - 021
C5 1.11740260e - 024
C6 -6.29238120e - 029
C7 0.00000000e + 000
C8 0.00000000e + 000
C9 0.00000000e + 000
Zernicke portion of the aspheric surface No. 21
ZER9 = 227.260 μm
ZER16 = 6.79061 μm
ZER25 = 1.17273 μm
ZER36 = 0.100691 μm
ZER49 = -0.00116301 μm
related to a half free diameter of 74.633 mm

[0039]

$$P(h) = \frac{\delta \cdot h \cdot h}{1 + \sqrt{1 - (1 + K) \cdot \delta \cdot \delta \cdot h \cdot h}} + \frac{C_1 h^4}{4} + \dots + C_n h^{2n+2}$$
$$\delta = 1 / R$$

[0040] where P is the sagitta as a function of the radius h (height to the optical axis 7), with the aspheric constants C, through C given in the Tables. R is the vertex radius given in the Tables.

TABLE 4

M1440a Lens	Radii	Thicknesses	Glasses	½ Lens Diameter	Refractive index at 351.1 nm
L 0	Infinity	35.0240	L710	60.887	.999982
L 1	-908.93348	7.0000	FK5	61.083	1.506235
	284.32550	6.4165	L710	63.625	.999982
L 2	968.84099	23.7611	FK5	64.139	1.506235
	-212.21935	.7000	L710	66.550	.999982
L 3	413.73094	17.2081	FK5	69.428	1.506235
	-424.88479	18.8724	L710	69.711	.999982
L 4	591.81336	19.7102	FK5	69.490	1.506235
	-250.67222	.7000	L710	69.228	.999982
L 5	-2772.23751	12.8582	FK5	67.060	1.506235
	-255.60433	.7000	L710	66.381	.999982
L 6	4699.63023	9.0382	FK5	62.603	1.506235
	120.65688	26.0302	L710	56.905	.999982

TABLE 4-continued

M1440a Lens	Radii	Thicknesses	Glasses	½ Lens Diameter	Refractive index at 351.1 nm
L 7	-182.28783	6.0000	FK5	56.589	1.506235
	302.39827	20.1533	L710	57.318	.999982
L 8	-140.55154	6.0000	FK5	57.674	1.506235
	205.78996	.7000	L710	64.913	.999982
L 9	197.09815	10.0000	FK5	66.049	1.506235
	223.79756	27.0961	L710	68.261	.999982
L 10	-191.72586	8.0000	FK5	70.299	1.506235
	-343.08973A	2.2458	L710	77.287	.999982
L 11	-292.95078	19.3593	FK5	77.813	1.506235
	-143.32621	.7000	L710	80.683	.999982
L 12	1440.49435	47.0689	FK5	95.650	1.506235
	-155.30867	.7000	L710	98.253	.999982
L 13	-2647.76343	13.8320	FK5	100.272	1.506235
	-483.82832	.7000	L710	100.543	.999982
L 14	169.62760	45.9417	FK5	99.308	1.506235
	-1090.68864	3.2649	L710	96.950	.999982
L 15	102.07790	10.0000	FK5	77.455	1.505235
	100.38160	40.1873	L710	73.370	.999982
L 16	-504.79995	6.0000	FK5	71.843	1.506235
	130.61081	34.6867	L710	64.992	.999922
L 17	-153.51955	6.0000	FK5	64.734	1.506235
	284.44035	34.2788	L710	67.573	.999982
L 18	-114.12583	8.2925	FK5	68.531	1.506235
	731.33965	20.4412	L710	84.132	.999982
L 19	-291.19603	24.2439	FK5	86.387	1.506235
	-173.68634	.7000	L710	93.185	.999982
L 20	-10453.06716	28.2387	FK5	111.655	1.506235
	-304.21017	.7000	L710	114.315	.999982
L 21	-2954.65846	30.7877	FK5	122.647	1.506235
	-312.03660	7.0000	L710	124.667	.999982
Diaphragm	Infinity	.0000		131.182	.999982
L 22	1325.30512	52.2352	FK5	133.384	1.506235
	-282.76663	.7000	L710	135.295	.999982
L 23	276.96510	52.6385	FK5	134.809	1.506235
	-1179.05517	25.2703	L710	132.935	.999982
L 24	-311.05526	10.0000	FK5	131.670	1.506235
	-587.25843	10.5026	L710	130.474	.999982
L 25	-374.19522	15.0000	FK5	130.116	1.506235
	-293.45628	.7000	L710	130.127	.999982
L 26	198.19004	29.6167	FK5	111.971	1.506235
	535.50347	.7000	L710	109.450	.999982
L 27	132.82366	34.0368	FK5	94.581	1.506235
	361.69797	12.8838	L710	90.620	.999982
L 28	7006.77771	9.7505	FK5	88.792	1.506235
	349.77435	1.0142	L710	79.218	.999982
L 29	174.38688	38.8434	FK5	73.443	1.506235
	55.37159	4.9107	L710	45.042	.999982
L 30	55.08813	42.8799	FK5	43.842	1.506235
	807.41351	1.9795	L710	30.725	.999982
	Infinity	3.0000	FK5	29.123	1.506235
	Infinity	12.0000		27.388	.999982

L710 is air at 950 mbar
ASPHERIC CONSTANTS of the image-side lens surface of the lens L10:
K = -0, 10137327 × 10²
C₁ = 0, 40139591 × 10⁻⁷
C₂ = 0, 011813452 × 10⁻¹²
C₃ = 0, 45259406 × 10⁻¹⁶
C₄ = 0, 29594697 × 10⁻²⁰
C₅ = 0, 11174026 × 10⁻²³
C₆ = -0, 62923812 × 10⁻²⁸

[0041] A lens arrangement 19 is shown in FIG. 5 with an image-side aperture of 0.7; it can be divided into six lens groups and consists only of spherical lenses. This embodiment example, in contrast to FIG. 4, has an extremely long first lens group, which comprises the lenses L1-L5. This elongate bulge is substantially formed by the thick positive lenses L4 and L5. A small distortion is attained with only spherical lenses by means of this first, elongate bulge, a worse input telecentricity being accepted, which is due to the

shape of this first bulge, and which can be compensated for by the illumination system. This first lens group has positive refractive power.

[0042] The second lens group L2 includes four negative lenses, a positive meniscus lens L9 which is curved toward the object again being arranged between the third negative lens L8 and the fourth negative lens L10. No aspheric lens surface is provided in this embodiment example. Astigma-

tism, coma and sagittal coma can in particular be corrected by means of this configuration of the first lens group LG2 with negative refractive power.

[0043] The third lens group comprises the lenses L11-L15 and has positive refractive power. In this embodiment example, in contrast to the first embodiment example, the lenses L12 and L14 are not so strongly pronounced. This third lens group has in particular a positive effect on the imaging quality in the quadrants.

[0044] The fourth lens group LG4 is formed, in spite of the high aperture of 0.70, by only three negative lenses, and thus has negative refractive power.

[0045] The adjoining fifth lens group LG5, which has positive refractive power, begins with the three positive lenses L19-L21, behind which the diaphragm is arranged. Behind the diaphragm are again arranged two thick positive lenses L22 and L23 which are constituted with strongly curved mutually facing lens surfaces. The focal lengths are f_{12} =486.1 and $f_{3,4}$ =431.429. The adjoining lenses L24 and L25 are provided for the correction of the oblique spherical aberration in the sagittal and tangential directions.

[0046] The sixth lens group comprises the lenses L28-L31 and has positive refractive power.

[0047] This objective has a numerical aperture of 0.7 at a wavelength of λ =351.4 nm The length from the image plane 0 to the object plane 0' is 1,000 mm, an image field of 8×26 mm being illuminated. All the lenses are made of crown glass, e.g., FK 5 of the SCHOTT Company. The objective requires laser light with a half value width of about 4.3 pm for the imaging of structures 210 nm wide, with a diagonal image field diameter of 27.20 mm. For a $\Delta\lambda$ of ± 2.15 pm, the longitudinal color error is ± 140 nm, and the transverse color error is at most 2.4 nm.

[0048] The exact lens data are given in Table 5.

TABLE 5

M1439a Lens	Radii	Thick- nesses	Glasses	½ Lens Diameter	Refractive index at 351.1 nm
0	Infinity	37.4632	L710	60.122	.999982
L1	-765.87757	6.1000	FK5	60.667	1.506235
	189.10818	15.0560	L710	63.489	.999982
L2	-1924.23712	14.8631	FK5	64.889	1.506235
	-210.10849	.7000	L710	66.477	.999982
L3	-1771.07074	12.9447	FK5	69.458	1.506235
	-265.44452	.7944	L710	70.441	.999982
L4	4429.34036	57.0511	FK5	72.296	1.506235
	-157.99293	10.8676	L710	76.220	.999982
L5	1697.66973	44.3818	FK5	72.396	1.506235
	-160.27202	3.1174	L710	69.891	.999982
L6	-270.20389	6.1000	FK5	64.496	1.506235
	145.59587	23.8094	L710	59.327	.999982
L7	-216.22267	6.1000	FK5	59.111	1.506235
	264.18566	22.2709	L710	60.114	.999982
L8	-154.67894	6.1000	FK5	60.693	1.506235
	682.06879	.7398	L710	66.605	.999982
L9	613.14630	9.9262	FK5	67.195	1.506235
	5830.30332	11.7064	L710	68.965	.999982
L10	-302.46010	6.1037	FK5	70.262	1.506235
	-1240.99707	11.3919	L710	74.655	.999982
L11	-304.61263	14.0742	FK5	76.744	1.506235
	-173.33791	.7386	L710	78.915	.999982
L12	-18363.19083	29.3863	FK5	86.930	1.506235
	-186.22288	.7000	L710	88.783	.999982

TABLE 5-continued

M1439a Lens	Radii	Thick- nesses	Glasses	½ Lens Diameter	Refractive index at 351.1 nm
L13	1172.01143	16.5023	FK5	92.295	1.506235
	-590.45415	.7000	L710	92.625	.999982
L14	157.32134	37.1074	FK5	92.978	1.506235
	6181.16889	.7000	L710	91.301	.999982
L15	146.43371	14.3174	FK5	82.500	1.506235
	202.92507	50.3624	L710	80.095	.999982
L16	-572.56205	6.1000	FK5	66.246	1.506235
	121.14512	30.5565	L710	59.998	.999982
L17	-157.40409	6.1000	FK5	59.747	1.506235
	256.95256	32.8427	L710	61.781	.999982
L18	-98.80342	7.8212	FK5	62.557	1.506235
	641.39965	18.1478	L710	77.492	.999982
L19	-308.76412	25.1716	FK5	80.101	1.506235
	-160.15183	.7198	L710	86.681	.999982
L20	-1043.39436	24.1973	FK5	100.363	1.506235
	-240.97724	.7732	L710	102.908	.999982
L21	1125.18771	28.8421	FK5	114.949	1.506235
	-430.47634	5.0000	L710	116.447	.999982
	Infinity	2.0000	L710	120.343	.999982
	Diaphragm	.0000		120.343	
L22	-3902.89359	42.5513	FK5	120.388	1.506235
	-230.34901	.7000	L710	122.380	.999982
L23	245.03767	46.6321	FK5	123.915	1.506235
	-1881.85244	29.1602	L710	122.327	.999982
L24	-267.61242	10.0000	FK5	121.023	1.506235
	-375.94475	6.6966	L710	120.815	.999982
L25	-319.48578	12.2841	FK5	120.185	1.506235
	-264.83170	.7000	L710	120.270	.999982
L26	212.78046	21.2016	FK5	105.054	1.506235
	451.72928	.7000	L710	103.107	.999982
L27	124.48521	34.0506	FK5	90.946	1.506235
	347.71817	14.6164	L710	87.431	.999982
L28	-10967.28804	15.6165	FK5	84.762	1.506235
	347.69074	.7000	L710	73.470	.999982
L29	175.92874	41.2004	FK5	69.063	1.506235
	53.54834	4.2189	L710	42.333	.999982
L30	53.63724	40.1656	FK5	41.450	1.506235
	1252.66340	3.3571	L710	30.075	.999982
L31	Infinity	3.0000	FK5	27.101	1.506235
	Infinity	12.0000	L710	25.514	.999982
	Infinity			13.603*	1.0

L710 is air at 950 mbar

I claim:

1. A projection objective comprising a lens arrangement comprising at least one lens group of negative refractive power, this lens group including at least four lenses of negative refractive power,
- wherein a lens of positive refractive power is arranged in this lens group after a third lens of negative refractive power.
2. The projection objective according to claim 1, wherein the lens of positive refractive power arranged in the lens group of negative refractive power comprises a meniscus lens.
3. The projection objective according to claim 1, wherein the lens of positive refractive power arranged in the lens group of negative refractive power comprises a convex lens surface.
4. The projection objective according to claim 1, wherein the projection objective comprises at least a second lens group of negative refractive power.
5. The projection objective according to claim 1, wherein the lens of positive refractive power is arranged in a first lens group of negative refractive power.

6. A double telecentric projection objective comprising an image side numerical aperture of at least 0.7 and a lens group in which a diaphragm is arranged, wherein lenses arranged before the diaphragm on a side facing toward a reticle have exclusively positive refractive power, a first lens of these lenses of positive refractive power having a positive focal length between 4 and 25 times a value of an object-to-image distance.

7. The double telecentric projection objective according to the preamble of claim 6, wherein a first lens of lenses of positive refractive power has a difference of radii that is smaller than 4% of an object-to-image distance 00'.

8. The double telecentric projection objective, comprising a first lens group of positive refractive power, a second lens group of negative refractive power, a third lens group of positive refractive power, and a fourth lens group of negative refractive power, wherein a respective air interspace is arranged between the second and the third lens groups, and the third and the fourth lens groups, a sum of these air spaces exceeding a sum of a lens thicknesses of the fourth lens group by at least 30%.

9. The double telecentric objective according to claim 8, wherein the air interspace between the third and the fourth

lens group comprises an extent in an axial direction that attains a value of at least 50% of a sum of glass thicknesses of the fourth lens group.

10. The double telecentric objective according to claim 8, wherein the air interspace between the third and the fourth lens group has an extent in an axial direction that attains a value of at least 60% of an extent of the air interspace between a second and the third lens group.

11. A projection exposure device for microlithography, comprising a projection objective comprising a lens arrangement according to claim 1.

12. A process for the manufacture of microstructured components, comprising the steps of exposing a substrate provided a photosensitive layer by means of ultraviolet laser light by means of a mask and

a projection exposure device with a lens arrangement according to claim 1, and structuring the substrate corresponding to a pattern contained on the mask, if necessary after development of the photosensitive layer.

* * * * *